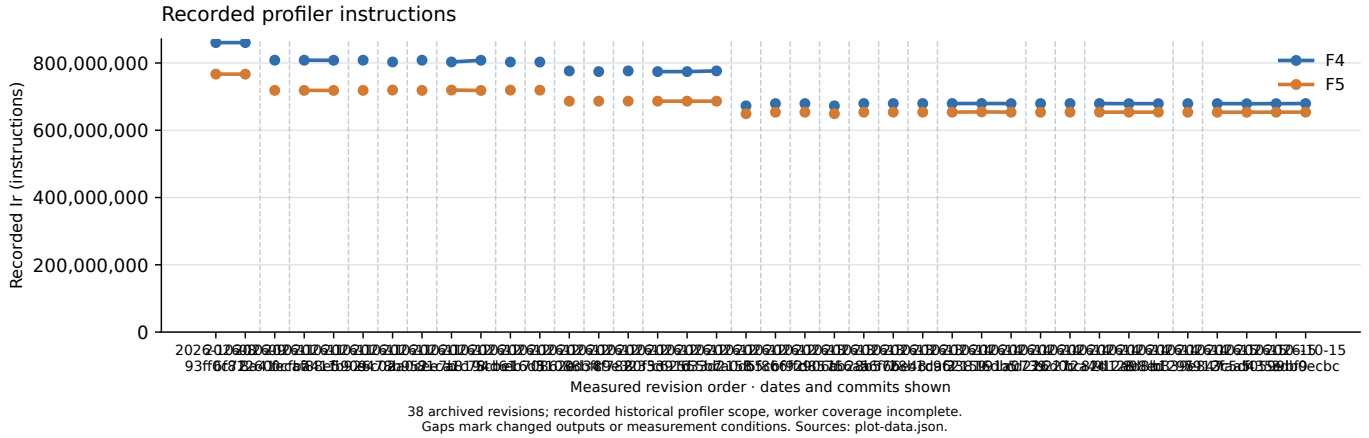
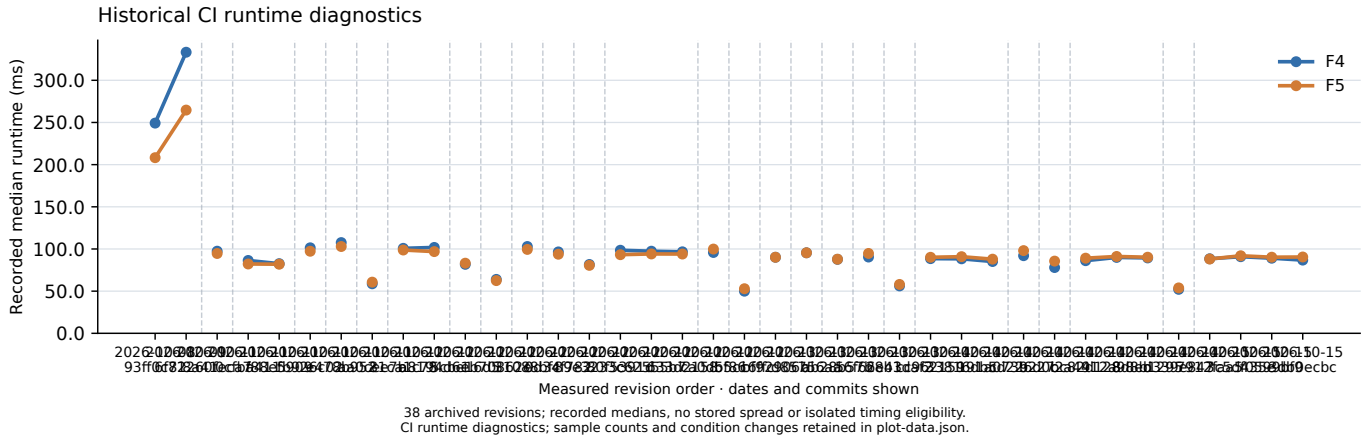
# Boolean matrix step: F4 and F5

One shared K_1 / 2^23 m=2 system, 22 Boolean variables, maximum degree 4. Matrix construction, reduction and readback; F5 includes its criterion.




| UTC date / revision | F4 Ir | F5 Ir | F4 ms | F5 ms |
|---|---|---|---|---|
| 2026-10-08 / 93ff0c72 | 860,633,889 | 766,940,469 | 249.241 | 208.250 |
| 2026-10-09 / 6f81260f | 860,555,567 | 766,776,087 | 333.279 | 264.687 |
| 2026-10-11 / 8a41eca7 | 808,263,705 | 718,576,235 | 97.380 | 94.658 |
| 2026-10-11 / 0cfba411 | 808,263,705 | 718,578,523 | 86.390 | 82.214 |
| 2026-10-11 / f88e5909 | 807,989,935 | 718,391,055 | 82.597 | 81.948 |
| 2026-10-11 / fb926c02 | 808,263,705 | 718,576,189 | 101.379 | 97.323 |
| 2026-10-11 / e478b952 | 803,017,657 | 719,516,063 | 107.585 | 102.927 |
| 2026-10-11 / aa0cee7b | 808,281,869 | 718,492,761 | 58.723 | 60.673 |
| 2026-10-12 / 81ca1c78 | 803,015,877 | 719,509,003 | 100.762 | 98.775 |
| 2026-10-12 / a8194d61 | 808,072,539 | 718,282,030 | 101.827 | 96.904 |
| 2026-10-12 / 5cbe16d5 | 802,824,065 | 719,316,911 | 81.784 | 83.144 |
| 2026-10-12 / eb70862e | 802,824,065 | 719,316,876 | 63.944 | 62.505 |
| 2026-10-12 / 31088bfc | 776,242,363 | 686,315,894 | 102.929 | 99.489 |
| 2026-10-12 / 0d34f782 | 774,379,185 | 686,521,131 | 96.521 | 93.790 |
| 2026-10-12 / 89e3233c | 776,490,367 | 686,543,371 | 81.628 | 80.581 |
| 2026-10-12 / 80f5391d | 774,379,185 | 686,519,896 | 98.501 | 93.239 |
| 2026-10-12 / c62555b7 | 774,379,185 | 686,519,896 | 97.360 | 94.117 |
| 2026-10-12 / 633d2158 | 776,467,358 | 686,512,000 | 96.637 | 94.008 |
| 2026-10-12 / ca0d5f86 | 672,500,776 | 649,174,334 | 95.848 | 99.895 |
| 2026-10-12 / b5cc6ffc | 679,242,389 | 653,622,849 | 49.965 | 52.882 |
| 2026-10-13 / bf92906b | 679,242,389 | 653,622,849 | 90.163 | 90.201 |
| 2026-10-13 / d85756aa | 672,486,486 | 649,160,075 | 95.497 | 95.480 |
| 2026-10-13 / ab28b57b | 679,449,939 | 653,823,121 | 87.697 | 87.878 |
| 2026-10-13 / 56f66e43 | 679,449,939 | 653,828,331 | 90.448 | 94.715 |
| 2026-10-13 / 28b1daf2 | 679,449,939 | 653,828,331 | 56.215 | 57.961 |
| 2026-10-14 / cc962159 | 679,449,939 | 653,831,656 | 88.546 | 90.149 |
| 2026-10-14 / 33816db5 | 679,549,997 | 654,627,509 | 88.344 | 90.874 |
| 2026-10-14 / 191a0739 | 679,240,609 | 653,664,759 | 85.135 | 87.917 |
| 2026-10-14 / cd226d0c | 679,240,609 | 653,664,759 | 92.059 | 98.073 |
| 2026-10-14 / b2272a44 | 679,449,939 | 653,873,401 | 78.007 | 85.621 |

| UTC date / revision | F4 Ir | F5 Ir | F4 ms | F5 ms |
|---|---|---|---|---|
| 2026-10-14 / bc82d128 | 679,449,939 | 653,873,401 | 86.126 | 89.102 |
| 2026-10-14 / 912a98ed | 679,098,623 | 653,830,968 | 89.995 | 91.125 |
| 2026-10-14 / 8d8b1295 | 679,098,623 | 653,830,968 | 89.395 | 90.230 |
| 2026-10-14 / b339e847 | 679,240,608 | 653,621,583 | 52.150 | 53.839 |
| 2026-10-15 / 7912faa5 | 679,226,318 | 653,606,027 | 88.406 | 88.091 |
| 2026-10-15 / 3c5d435e | 678,905,208 | 653,631,022 | 90.855 | 91.956 |
| 2026-10-15 / f0599bf9 | 679,098,622 | 653,825,706 | 89.060 | 90.253 |
| 2026-10-15 / 3db0ecbc | 679,449,938 | 653,830,913 | 86.813 | 90.492 |

Historical Python-orchestrated perf.history/v1 snapshots; no new performance run or speedup claim.

Runtime: one recorded median per kernel and revision; sample counts retained per point, no stored spread, no isolated timing eligibility. Runtime is diagnostic.

Profiler Ir is the recorded historical scope. Worker-thread coverage is incomplete for these kernels; it is not a complete solver operation count.

Lines stop at missing values, changed output fingerprints, or changed host/compiler/harness/profiler. Different algorithm fingerprints are expected and are not equated.

X positions are measured revision order, labelled with UTC commit date and revision; spacing is not elapsed time.

Evidence: 38 imported perf.history/v1 snapshots; historical_sources in plot-data.json retains raw filenames, SHA-256 hashes, hosts, harnesses and sample counts. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.